



DWF_dp_sign_select functions

Sign Selection / Conditional Two's Complement

Version, STAR, and myDesignWare Subscriptions: [IP Directory](#)

Description

Revision History

The DWF_dp_sign_select functions return the positive or negative (two's complement) value of argument *a* controlled by sign argument *s*. Argument *a* and the return value are both either signed (two's complement) or unsigned.

A signed return value has overflow for the value $\text{DWF_dp_sign_select}(-2^{\text{width}-1}, 1) = 2^{\text{width}-1}$, which cannot be represented as a signed number of *width* bits. The complement of an unsigned number always results in underflow (that is, a negative number cannot be represented as an unsigned), but the unsigned DWF_dp_sign_select can be meaningful for the conditional addition/subtraction in a larger unsigned expression.

Table 1-1 Function Names

Function Name	Description
DWF_dp_sign_select	VHDL unsigned sign select
DWF_dp_sign_select	VHDL signed (two's complement) sign select
DWF_dp_sign_select_uns	Verilog unsigned sign select
DWF_dp_sign_select_tc	Verilog signed (two's complement) sign select

Table 1-2 Argument Description

Argument Name	Type	Width	Description
<i>a</i>	Vector	<i>width</i>	Input data
<i>s</i>	Bit	1	Sign / complement control
DWF_dp_sign_select	Vector	<i>width</i>	Return value

Table 1-3 Parameter Description (Verilog)

Parameter	Values	Description
<i>width</i>	≥ 1	Word length of input <i>a</i> and return value

Verilog Include File: DW_dp_sign_select_function.inc

Functional Description

```
z[width-1:0] = DWF_dp_sign_select (a[width-1:0], s)
z[width-1:0] =  a[width-1:0]  if s == 0
                = -a[width-1:0] else
```

For more information about the DesignWare datapath functions, refer to the topic titled [Arithmetic - Datapath Functions Overview](#).

Related Topics

- [Arithmetic - Datapath Functions Overview](#)
- [DesignWare Building Block IP User Guide](#)

VHDL Example

```

library IEEE, DWARE;
use IEEE.std_logic_1164.all;
use IEEE.numeric_std.all;
use DWARE.DW_dp_functions.all;
-- DWARE.DW_dp_functions_arith package if IEEE.std_logic_arith is used

entity DWF_dp_sign_select_test is
  port (a, b, c : in  signed(7 downto 0);
        s       : in  std_logic;
        z       : out signed(15 downto 0));
end DWF_dp_sign_select_test;

architecture rtl of DWF_dp_sign_select_test is
begin
  z <= DWF_dp_sign_select (a * b, s) + c;
end rtl;

```

Verilog Example

```

module DWF_dp_sign_select_test (a, b, c, s, z);

  input  signed [7:0] a, b, c;
  input          s;
  output signed [15:0] z;

  // Passes the parameter to the function
  parameter width = 16;

  // add "$SYNOPSISYS/dw/sim_ver" to the search path for simulation
  `include "DW_dp_sign_select_function.inc"

  assign z = DWF_dp_sign_select_tc (a * b, s) + c;

endmodule

```

Revision History

For notes about this release, see the [DesignWare Building Block IP Release Notes](#).

For lists of both known and fixed issues for this component, refer to the [STAR report](#).

Date	Release	Updates
September 2024	DWBB_202409.0	■ Updated date and version; no technical changes
December 2023	DWBB_202312.0	■ Updated date and version; no technical changes ■ Added this Revision History table and the document links on this page

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